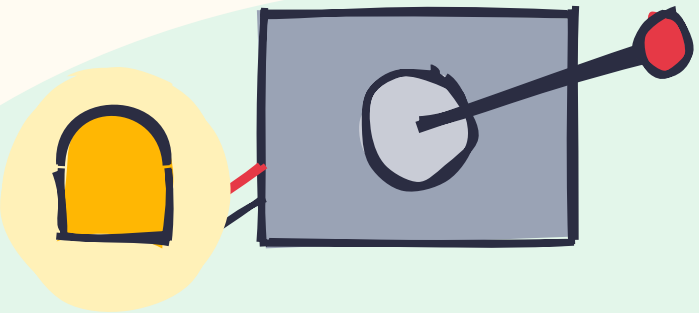
Little Explorers · Class 24

Crank-Powered Light

You are the power station today!

Moving = power | Light an LED | Save energy



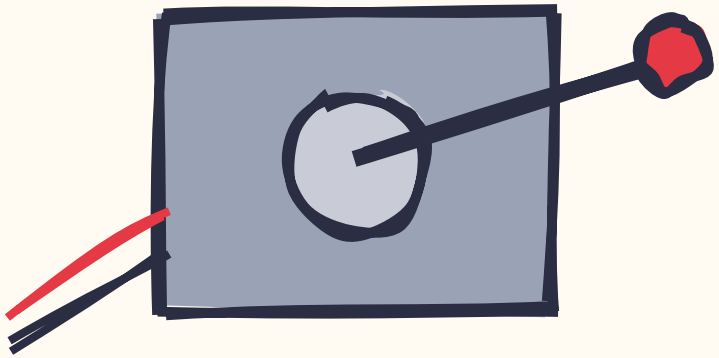
Did you know?
Some emergency radios and torches have a crank. Turn the handle and they work without batteries!

Let's wonder...

Where does electricity come from?

Load-shedding!

When the lights go out at home, everything stops. Where does electricity come from in the first place? Could **you** make some with your own hands?



From the wall?

From batteries?

From spinning!

Today: you turn a **crank generator** with your hand and light an LED: you are the power station!

Learning objectives

Today I can…

1	**Moving makes power:** say that **spinning a generator** makes electricity	**Proof:** I explained what my turning hand does
2	**Light it up:** turn a crank to **light an LED**	**Proof:** I lit the LED by turning
3	**Save energy:** name ways to make and **save** electricity	**Proof:** I said one way to save energy at home

Objective 1 · Moving makes power · video

Video time!



How is Electricity made?

Maddie's Do You Know? · about 6 minutes · youtu.be/X8iV_TkkbBg

Watch for...

1. Where does electricity come **from**?
2. What spins inside a power station?
3. Name a **clean** way to make electricity.

Scan to watch

After the video: Rub your hands together fast. Do they get warm? Moving makes energy!

Objective 1 · Moving makes power

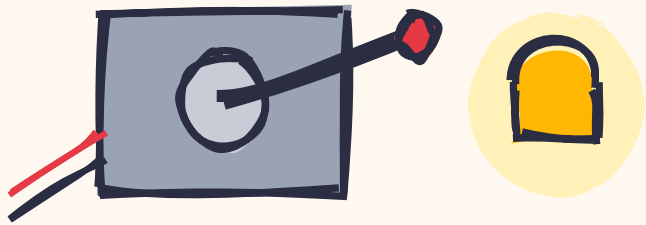
Spinning makes electricity

The idea

Inside a **generator** are **magnets** and **wire**. When something spins them, electricity flows!

Big power stations spin giant generators with **water**, **wind** or **steam**. Today, **your hand** does the spinning.

Example: a bicycle light



Some bikes have a little dynamo on the wheel. Pedal fast: the light shines bright. Stop: the light goes out. Your legs are making electricity!

Objective 2 · Light it up

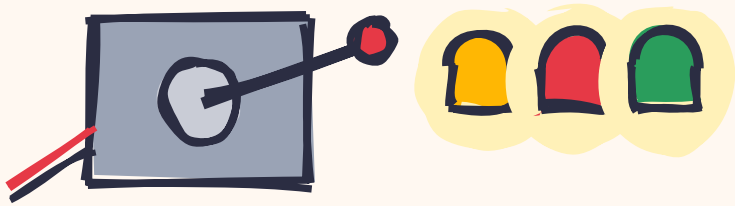
Harder work for more lights

The idea

Try lighting **2** or **3** LEDs, or a little buzzer. Does it feel **harder** to turn?

More things to power = more work for the power station. That's why power stations need lots of energy!

Example: carrying school bags



Carrying one bag is easy. Carrying three bags is hard work! The generator 'carries' each light, so more lights means harder turning.

Objective 3 · Save energy

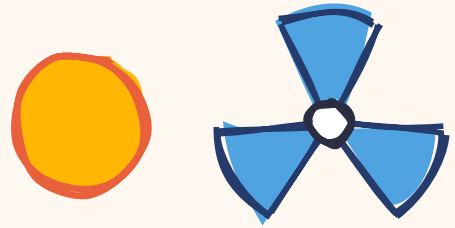
Clean power and saving energy

The idea

Wind, **water** and **sunshine** make clean electricity with no smoke.

We can all **save** energy: switch off lights and fans when we leave a room, and unplug chargers!

Example: turning off a tap

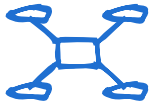


sun wind

If you leave a tap running, water is wasted. Leaving lights on wastes electricity the same way. Switch it off!

Real world

Where electricity is made



Dams
Tarbela Dam spins generators with river water



Solar panels
Sunlight makes electricity on rooftops



Wind farms
Windmills in Sindh spin in the wind



Crank torches
Your hand makes the power!



Which way of making electricity is cleanest? Why?

Tarbela
Tarbela Dam is one of the biggest dams in the world, and it makes lots of Pakistan's electricity!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human power station
2. Be the power station
3. Slow or fast?
4. Power more things
5. Power challenges

Class game

Human power station

How to play

1. Stand in a circle holding hands: you are the wire!
2. One child in the middle spins round: the generator.
3. When the generator spins, squeeze hands round the circle: the electricity flowing!
4. When it stops, the squeezing stops.

Then discuss

What happened to the 'electricity' when the generator stopped spinning?

Big idea

No spinning, no electricity. Power stations must keep spinning!



MAKE

Objective 2 · Build step 1 of 2

Be the power station

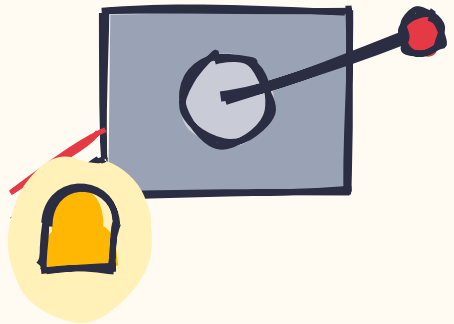
a Hold the crank generator steady on the table.

b Check the LED is clipped on (long leg to red).

c Turn the handle slowly. What happens?

d Now turn it fast!

e Stop. What happens to the light?



Check: The LED lights up when you turn, and goes off when you stop.

Turn gently. Very fast turning can break the gears.

Experiment!

Slow or fast?

How I turn	I think it will be…	What happened
Turn slowly	dim / bright?	_____
Turn fast	dim / bright?	_____
Turn the other way	light / no light?	_____

Talk about it

Did turning the other way work? (An LED only lets electricity go ONE way!)

One way only

LEDs are like a one-way door for electricity.

MAKE

Objective 2 · Build step 2 of 2

Power more things

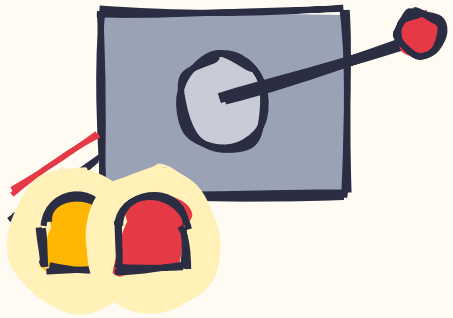
a Add a second LED next to the first.

b Turn the crank. Is it harder?

c Try a small buzzer instead.

d Take turns: how long can you keep the light on?

e Count to 10 while turning. Are your arms tired?



Check: More lights feel harder to turn.

Level up!

Power challenges

Choose your challenge

- **Stopwatch:** how long can your team keep the light on, taking turns?
- **Brightest:** who can make it brightest without breaking it?
- **Energy hunt:** find 3 things in school that use electricity.
- **Save it:** make a 'Switch it off!' poster.

Think like an engineer

Why do power stations need so much water, wind or steam?

Boss level

Light the LED with your partner turning while you hold the wires.

Objective check

Can I really do it? Prove it.

01	What does spinning a generator make?	○ ○ ○
02	Show your LED lighting up when you turn.	○ ○ ○
03	Tell us one way to save electricity at home.	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Power station relay: teams take turns keeping the light on for 1 minute.

Show what you know

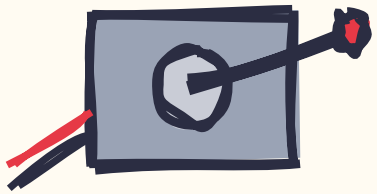
What did we learn?

- [x] Spinning magnets and wire make electricity
- [x] Faster turning = brighter light
- [x] More lights = harder work
- [x] Clean energy: sun, wind and water
- [x] Switch it off to save energy!

Exit ticket: what happens to the light when you stop turning?

At home

Count how many lights and fans are switched on at home. Can you switch some off?



Next class: My Talking Pet in Scratch!